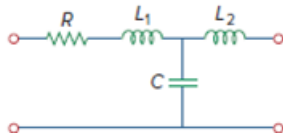


Problem

Using Fig. 19.90, design a problem to help other students better understand how to find the h and g parameters for a circuit in the s -domain.

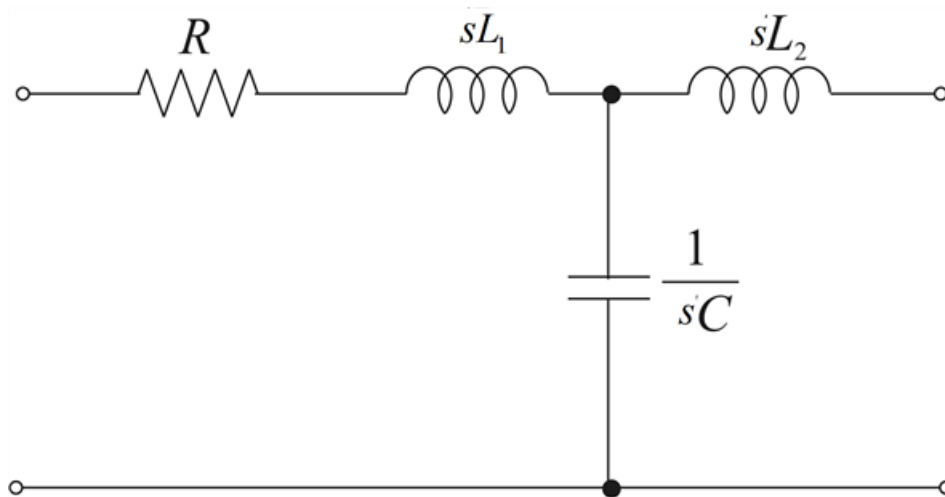
Figure 19.90



Step-by-step solution

Step 1 of 13

Given circuit in S – domain



Step 2 of 13

Hybrid parameters:-

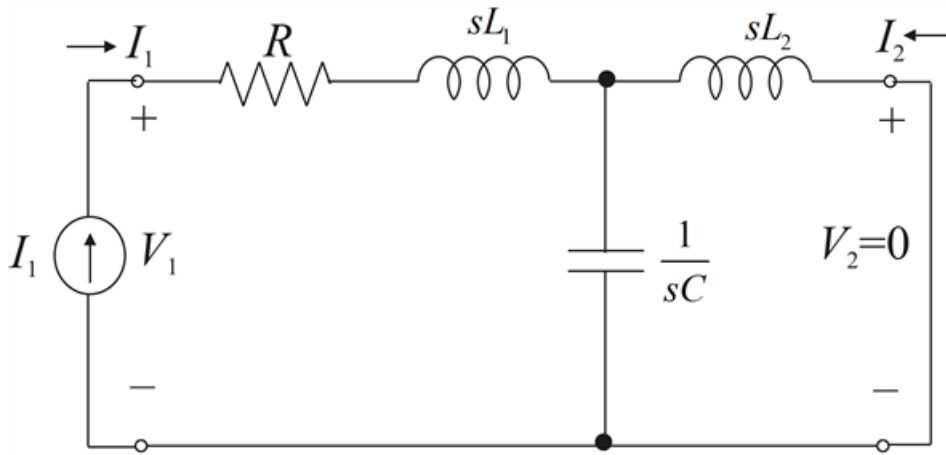
General form of equations are

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

To find h_{11} and h_{21} we short circuit the output port and connect a current source I_1 to the input port

Step 3 of 13



Step 4 of 13

sL_2 $\frac{1}{sC}$ are in parallel and in series with R, sL_1 ,

$$\left(sL_2 \parallel \frac{1}{sC} \right) + R + sL_1$$

$$\frac{(sL_2) \left(\frac{1}{sC} \right)}{sL_2 + \frac{1}{sC}} + R + sL_1$$

$$\Rightarrow \frac{sL_2}{s^2 L_2 C + 1} + R + sL_1$$

Input voltage is

$$V_1 = \left[\frac{sL_2 + s^2 L_2 RC + R + s^3 L_1 L_2 C + sL_1}{s^2 L_2 C + 1} \right] I_1$$

$$\frac{V_1}{I_1} = \frac{s^3 L_1 L_2 C + s^2 RL_2 C + s(L_1 + L_2) + R}{s^2 L_2 C + 1} \dots\dots\dots (1)$$

We know that

$$h_{11} = \frac{V_1}{I_1} \bigg|_{V_2=0}$$

$$\therefore h_{11} = \frac{s^2 L_1 L_2 C + s^2 RL_2 C + s(L_1 + L_2) + R}{s^2 L_2 C + 1}$$

Step 5 of 13

Using current division, we get

$$I_2 = \left(\frac{\frac{-1}{sC}}{\frac{1}{sC} + sL_2} \right) I_1$$

$$\frac{I_2}{I_1} = \frac{-1}{1 + s^2 L_2 C}$$

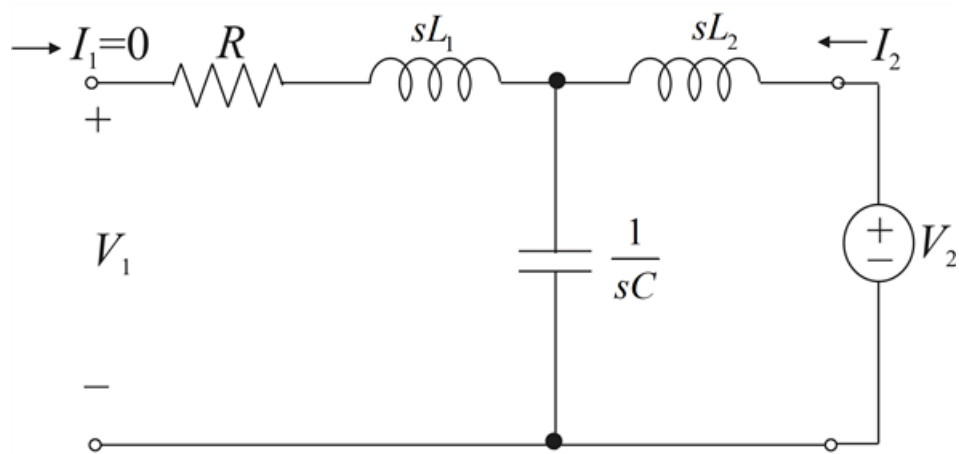
We know that

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$$

$$\therefore h_{21} = \frac{-1}{1 + s^2 L_2 C}$$

Step 6 of 13

To get h_{12} and h_{22} , we open-circuit the input port and connect a voltage source V_2 to the output port.



By voltage division, we get

$$V_1 = \left(\frac{\frac{1}{sC}}{\frac{1}{sC} + sL_2} \right) V_2$$

$$\frac{V_1}{V_2} = \frac{1}{1 + s^2 L_2 C}$$

We know that

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0}$$

$$\therefore h_{12} = \frac{1}{1 + s^2 L_2 C}$$

Step 7 of 13

Using KVL around V_2 , sL_2 , $\frac{1}{sC}$ then

$$-V_2 + sL_2 I_2 + \frac{1}{sC} I_2 = 0$$

$$I_2 \left(sL_2 + \frac{1}{sC} \right) = V_2$$

$$\frac{I_2}{V_2} = \left[\frac{sC}{s^2 L_2 C + 1} \right]$$

We know that

$$h_{22} = \frac{I_2}{V_2} \Big|_{I_1=0}$$

$$\therefore h_{22} = \left(\frac{sC}{s^2 L_2 C + 1} \right)$$

Thus, the hybrid parameters in matrix form are

$$[h] = \begin{bmatrix} \left(\frac{s^3 L_1 L_2 C + s^2 R L_2 + s(L_1 + L_2) + R}{s^2 L_2 C + 1} \right) & \frac{1}{s^2 L_2 C + 1} \\ \frac{-1}{s^2 L_2 C + 1} & \left(\frac{sC}{s^2 L_2 C + 1} \right) \end{bmatrix}$$

Step 8 of 13

Inverse hybrid parameters of g parameters:-

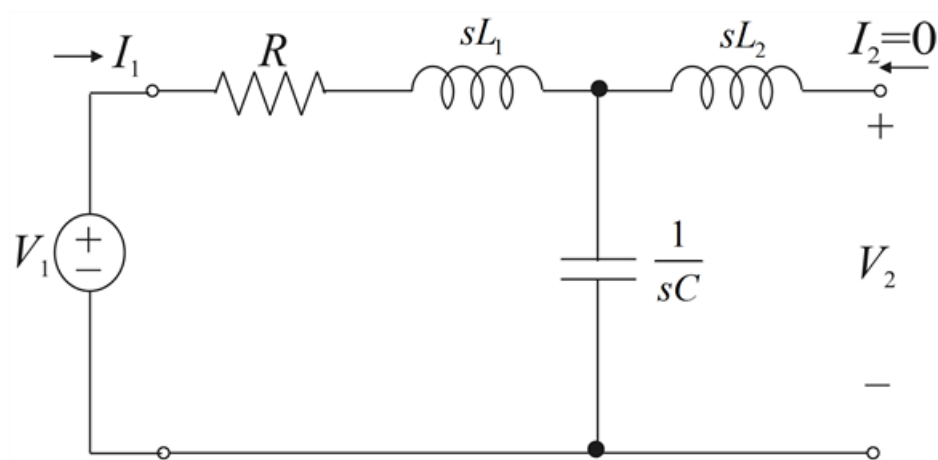
General equations are

$$I_1 = g_{11} V_1 + g_{12} I_2$$

$$V_2 = g_{21} V_1 + g_{22} I_2$$

To get g_{11} and g_{21} , we open-circuit the output port and port and connect a voltage source V_1 to the input port

Step 9 of 13



Step 10 of 13

The input current is

$$I_1 = \frac{V_1}{R + sL_1 + \frac{1}{sC}}$$

$$\frac{I_1}{V_1} = \frac{sC}{s^2L_1C + sRC + 1}$$

We know that

$$g_{11} = \left. \frac{I_1}{V_1} \right|_{I_2=0}$$

$$\therefore g_{11} = \left(\frac{sC}{s^2L_1C + sRC + 1} \right)$$

Step 11 of 13

By voltage division

$$V_2 = \left(\frac{\frac{1}{sC}}{R + sL_1 + \frac{1}{sC}} \right) V_1$$

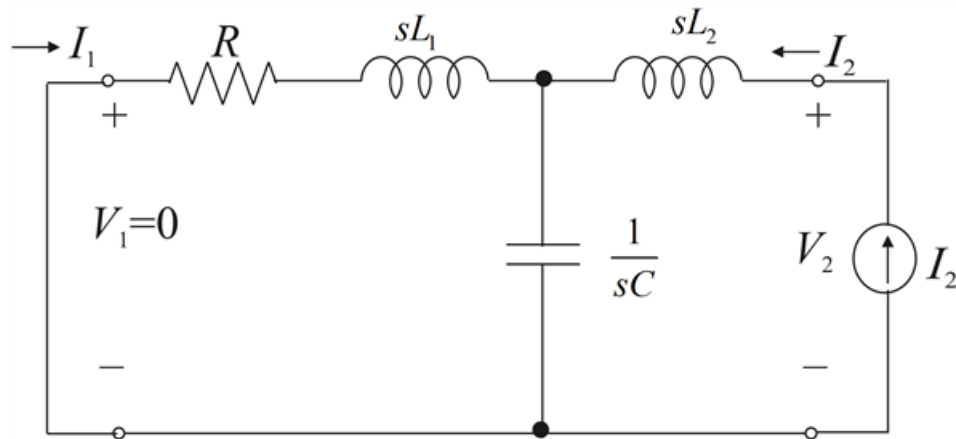
$$\frac{V_2}{V_1} = \frac{1}{s^2L_1C + sRC + 1}$$

We know that

$$g_{21} = \left. \frac{V_2}{V_1} \right|_{I_2=0}$$

$$\therefore g_{21} = \frac{1}{s^2L_1C + sRC + 1}$$

To get g_{12} and g_{22} we short circuit the input port and connect current source I_2 to the output port.



Step 12 of 13

By current division, we get

$$I_1 = \left(\frac{\frac{-1}{sC}}{\frac{1}{sC} + R + sL_1} \right) I_2$$

$$\frac{I_1}{I_2} = \frac{-1}{s^2 L_1 C + sRC + 1}$$

We know that

$$g_{12} = \frac{I_1}{I_2} \Big|_{V_1=0}$$

$$\therefore g_{12} = \frac{-1}{s^2 L_1 C + sRC + 1}$$

Step 13 of 13

Where R, sL_1 , are series in parallel to $\frac{1}{sC}$

$$\begin{aligned} & (R + sL_1) \parallel \frac{1}{sC} \\ \Rightarrow & \frac{(R + sL_1) \left(\frac{1}{sC} \right)}{R + sL_1 + \frac{1}{sC}} \\ \Rightarrow & \frac{R + sL_1}{s^2 L_1 C + sRC + 1} \text{ in series with } sL_2 \\ \Rightarrow & \frac{R + sL_1}{s^2 L_1 C + sRC + 1} + sL_2 \end{aligned}$$

Thus, the output voltage is

$$\begin{aligned} V_2 &= \left[\frac{R + sL_1}{s^2 L_1 C + sRC + 1} + sL_2 \right] I_2 \\ \frac{V_2}{I_2} &= \frac{R + sL_1}{s^2 L_1 C + sRC + 1} + sL_2 \end{aligned}$$

We know that

$$g_{22} = \frac{V_2}{I_2} \Big|_{V_1=0}$$

$$\therefore g_{22} = \frac{R + sL_1}{s^2 L_1 C + sRC + 1} + sL_2$$

Therefore, g parameters are in matrix form are.

$$[g] = \begin{bmatrix} \frac{sC}{s^2 L_1 C + sRC + 1} & \frac{-1}{s^2 L_1 C + sRC + 1} \\ \frac{1}{s^2 L_1 C + sRC + 1} & \frac{R + sL_1}{s^2 L_1 C + sRC + 1} + sL_2 \end{bmatrix}$$